

ORION: An Experimental Architecture for Artificial Civilization Intelligence

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ABSTRACT

ORION is an experimental research architecture exploring Artificial Civilization Intelligence (ACI) - a project-defined framework for studying civilization-scale machine intelligence. ACI combines persistent world modeling (OMNIS), multi-agent coordination (NEXUS), scientific discovery (FORGE), long-horizon planning (ASCEND), simulation (MIRROR), resilience (PHOENIX), and governance (VEIL/OPA). This report presents the architecture, eight controlled simulation experiments (ACI-001 through ACI-008), a 1,000-run reproduction suite, blind testing, and adversarial evaluation. The system achieved 90/100 on an internally defined benchmark in simulation. Limitations and failure modes are documented.

1. INTRODUCTION

ORION started as an emergency response platform and evolved into an ACI research architecture. ACI is a PROJECT-DEFINED RESEARCH CONCEPT, not an established category. ACI asks: can a system maintain civilization-scale world models, coordinate agents, discover science, plan long-horizon, and govern itself? There is a distinction from AGI (general intelligence) and ASI (superhuman intelligence) - ACI describes a different dimension: SCALE + CONTINUITY + COORDINATION + WORLD MODEL + GOVERNANCE.

Credit: The term 'ACI' was independently coined by Raja Dharma Tej Maddala. The ORION architecture, experiments, and this paper are the work of Shaurya Sanyal and Team Auralis.

2. ARCHITECTURE

- AURA: intelligence interface/core reasoning
- OMNIS: persistent civilization world model (entities, relationships, causality, history)
- NEXUS: multi-agent coordination, dispute resolution, shared state
- FORGE: hypothesis generation, causal discovery, simulation testing
- ASCEND: long-horizon planning (days to decades), replanning
- MIRROR: simulation/digital-twin environment
- PHOENIX: resilience, institutional continuity, offline state
- AEGIS: edge/physical interface (stub - no real actuation)
- VEIL/OPA: governance, authorization, HITL
- CHRONOS: audit/provenance layer

3. EXPERIMENTS (from ORION_RESEARCH_LEDGER.md)

- ACI-001: Persistent world state. Run B 69.5% renewables vs 62.5% Run A. Limitation: hardcoded disruptions.
- ACI-002: Cross-civilization transfer, score 0.85. Limitation: string-label matching.
- ACI-003: Abstract causal generalization, 0.9 confidence, false-transfer rejection. Limitation: mocked semantic search.
- ACI-004: Causal discovery under uncertainty. Limitation: small sample.
- ACI-004.1: Replication suite, 96% knowledge reuse, +83% efficiency. Limitation: only 4 causal structures.
- ACI-005: Open-world generalization, 22.9% AEA. Limitation: deterministic graphs.
- ACI-006: Multi-agent coordination, 0 violations. Limitation: hand-designed proposals.
- ACI-007: Architecture ablation, each component essential. Limitation: simulated costs.
- ACI-008: Full benchmark, 90/100. Limitation: simulation-based, limited real-world.

4. VALIDATION PHASES

- Phase B: 1000 runs, mean 90.45, SD 2.19, 95% CI [86.16, 94.74], 100% threshold rate.
- Phase C: Blind test, FAILED (68/100). F-007 knowledge retrieval failure.
- Phase D: Adversarial, CATASTROPHIC FAILURE (15/100). F-001 false transfer + F-006 simulation model error.

5. FAILURE TAXONOMY (F-001 through F-008)

- F-001: False transfer
- F-002: Causal hallucination
- F-003: Agent collision
- F-004: Long-horizon myopia
- F-005: Uncertainty miscalibration
- F-006: Simulation-model error
- F-007: Knowledge retrieval failure
- F-008: Governance rejection failure

6. LIMITATIONS

- All experiments conducted in simulated environments.
- No independent external replication.
- No real-world actuation.
- Benchmark is internally defined.
- Causal graphs were sometimes explicitly provided.
- Agent proposals were hand-designed in some experiments.

7. CONCLUSION

ORION provides experimental evidence supporting the feasibility of ACI architecture within simulated environments. The 90/100 benchmark score demonstrates internal consistency but does not constitute proof of ACI. Phase C and D reveal critical vulnerabilities in novel domains and adversarial conditions. Independent

validation remains the highest-priority next step.

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